Provor : a range of profiling floats for Operational Oceanography

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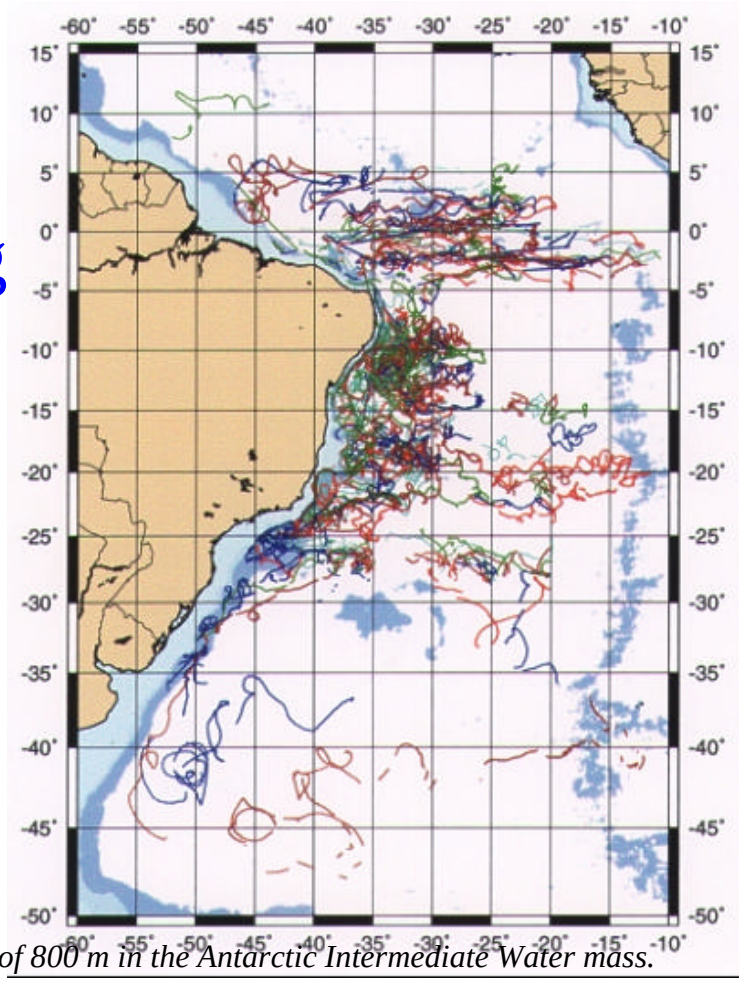
Provor : the Marvor heritage

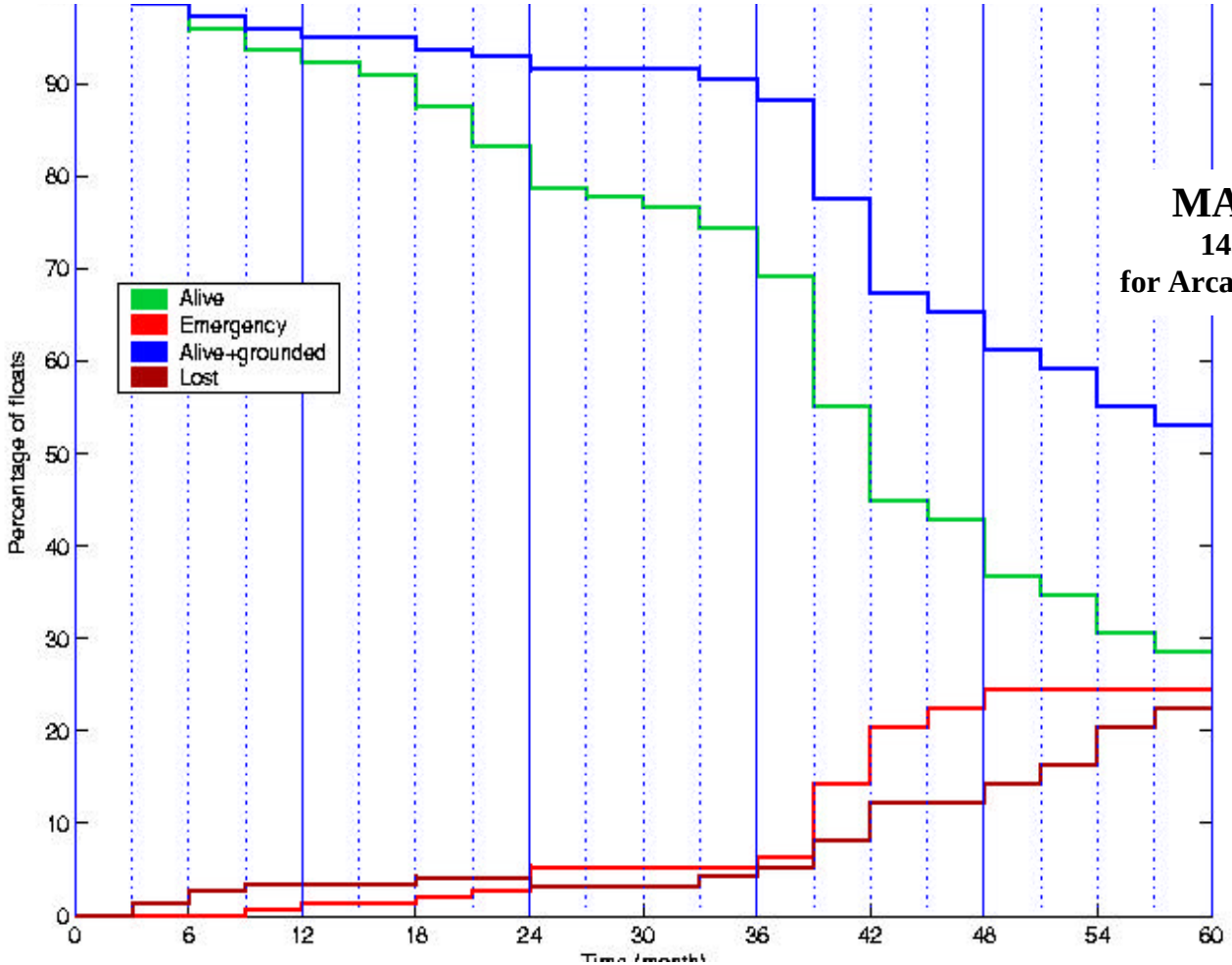
Based on experience
gained in developing & operating
multi-cycles MARVOR floats

➔ A 7 year at sea experience
➔ ~200 MARVOR floats deployed
➔ reliable, long life



Currents at a depth of 800 m in the Antarctic Intermediate Water mass.
Nearly 100 Marvor floats have been launched during the SAMBA experiment in the Brazil Basin between 1994 and 1997 for a 5 year mission

MARVOR results: 65% still alive after 3 years



MARVOR floats statistics

143 MARVOR floats launched

for Arcane, Eurofloat, Samba experiments

Provor - main features

- FSI or SeaBird CTD packages
- Self-ballasted, with active buoyancy control (hydraulic pump)
- Max profiling depth : 2000 m
- 100 to 150 cycles
- Possibility of data acquisition on down-profile to parking depth
- Great versatility in vertical sampling strategy (upper and deep layers with different resolution)
- ARGOS data transmission

Provor range of floats

Provor T
Seascan

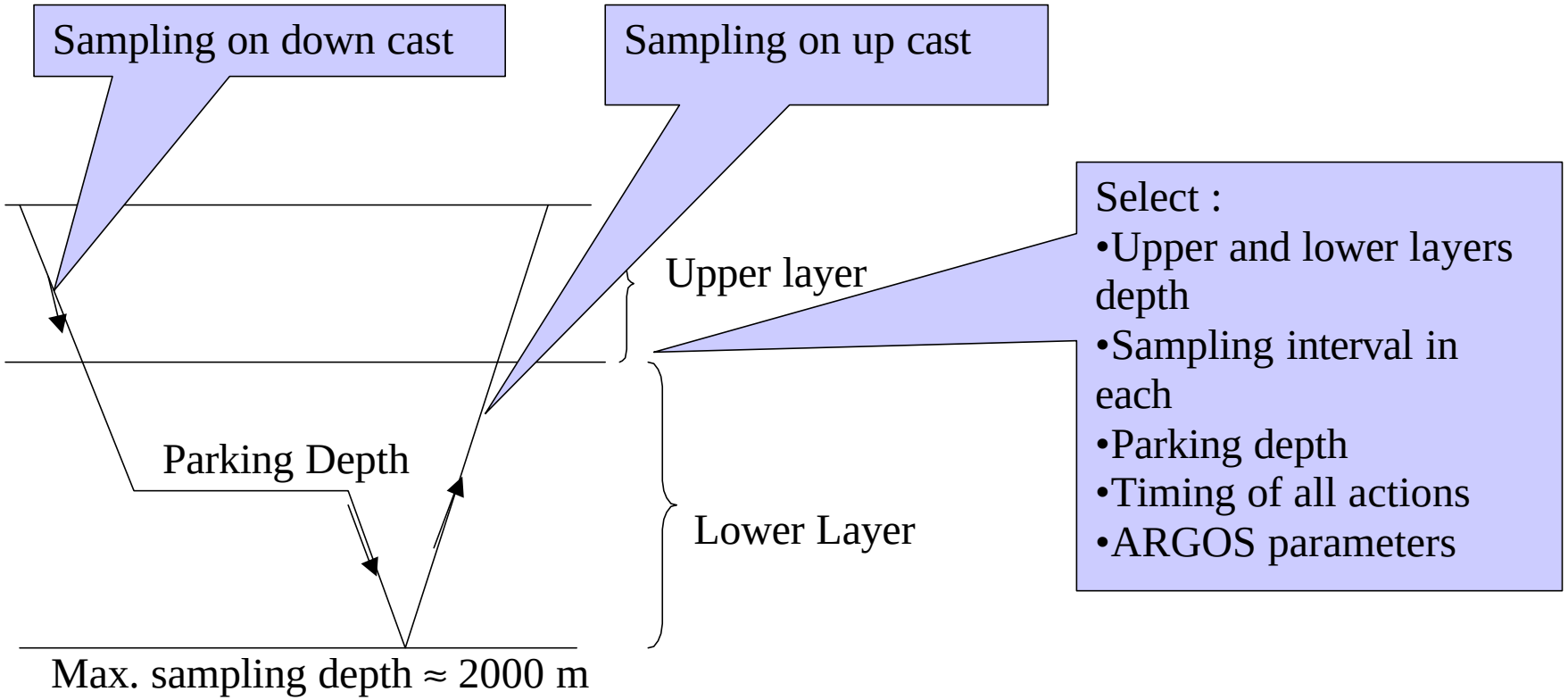
Provor CT-F2
FSI Excell

Provor-CT-S
Seabird 41 CP

Metocean Provor CT-S

- SBE sensor
- ready to deploy (removable magnet)
- 150 cycles (36 lithium D-cells)

Provor mission programme



Buoyancy control

- Transfer of oil between an internal reservoir and an external ballast
- No ballasting operation before launching
- +/- 30 dbars at depth
- About 2.3 liters of usable oil:
  - emergence at the surface : >1 liter
  - from surface to depth : 350 cm3 (function of difference of density)
  - quality of machining : 500 cm3

Data Acquisition

	P	T	C/S	Reduction
Provor-T	1 dbar	0.01 °C	-	Chords
Provor-CT-F2	1 dbar	0.01 °C	0.01mS/cm	Slices
Provor-CT-S	1 dbar	0.002 °C	0.005 PSU	Slices

- One sample every 10 seconds (1 sample per meter).
  No power switching to avoid electrical noise

- From 2000 points to 100 to reduce the time at the surface
  (using Argos data transmission system)

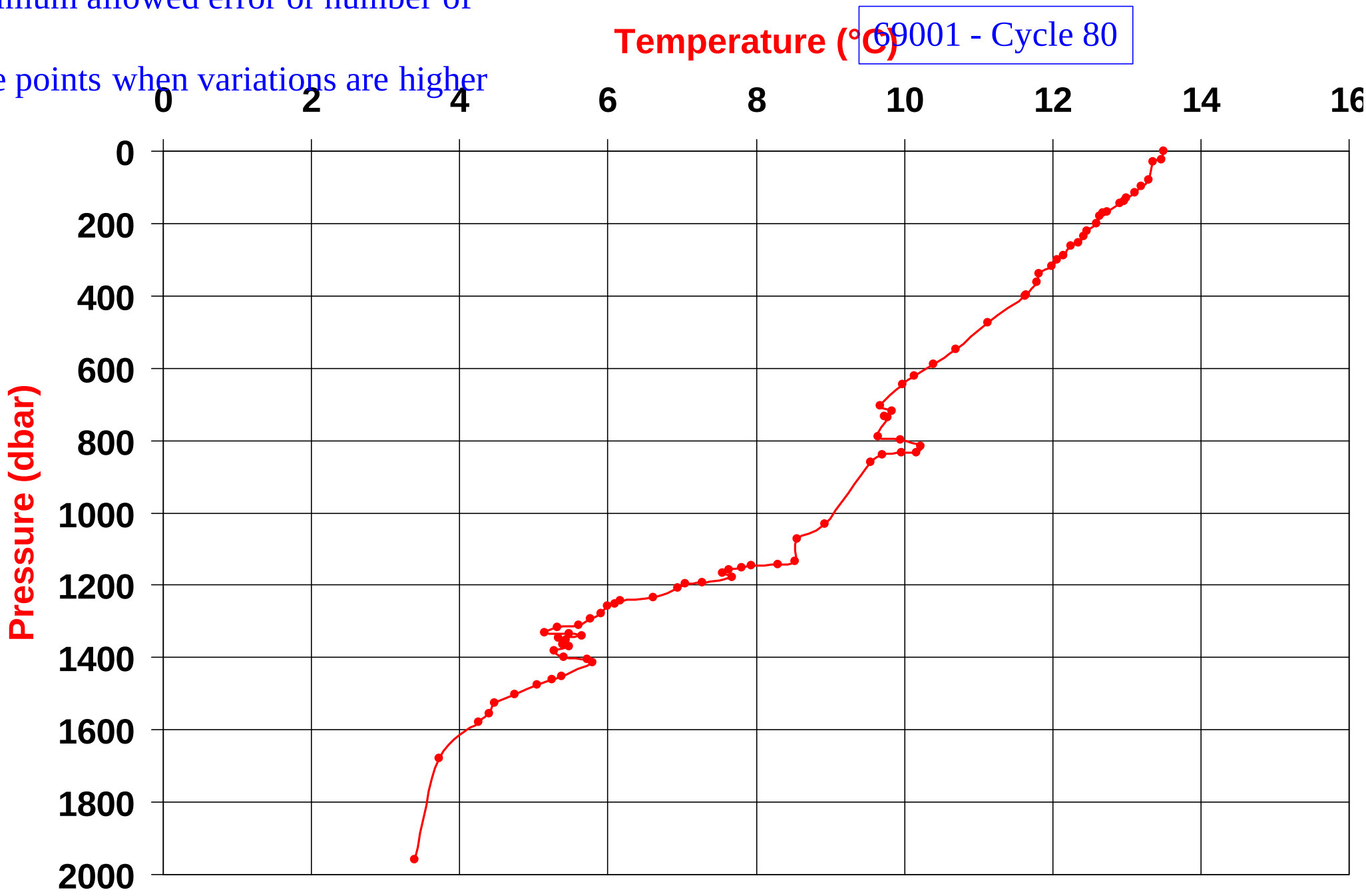
- 2 data reduction methods : chords or slices

Data reduction : chord

Segments between real points

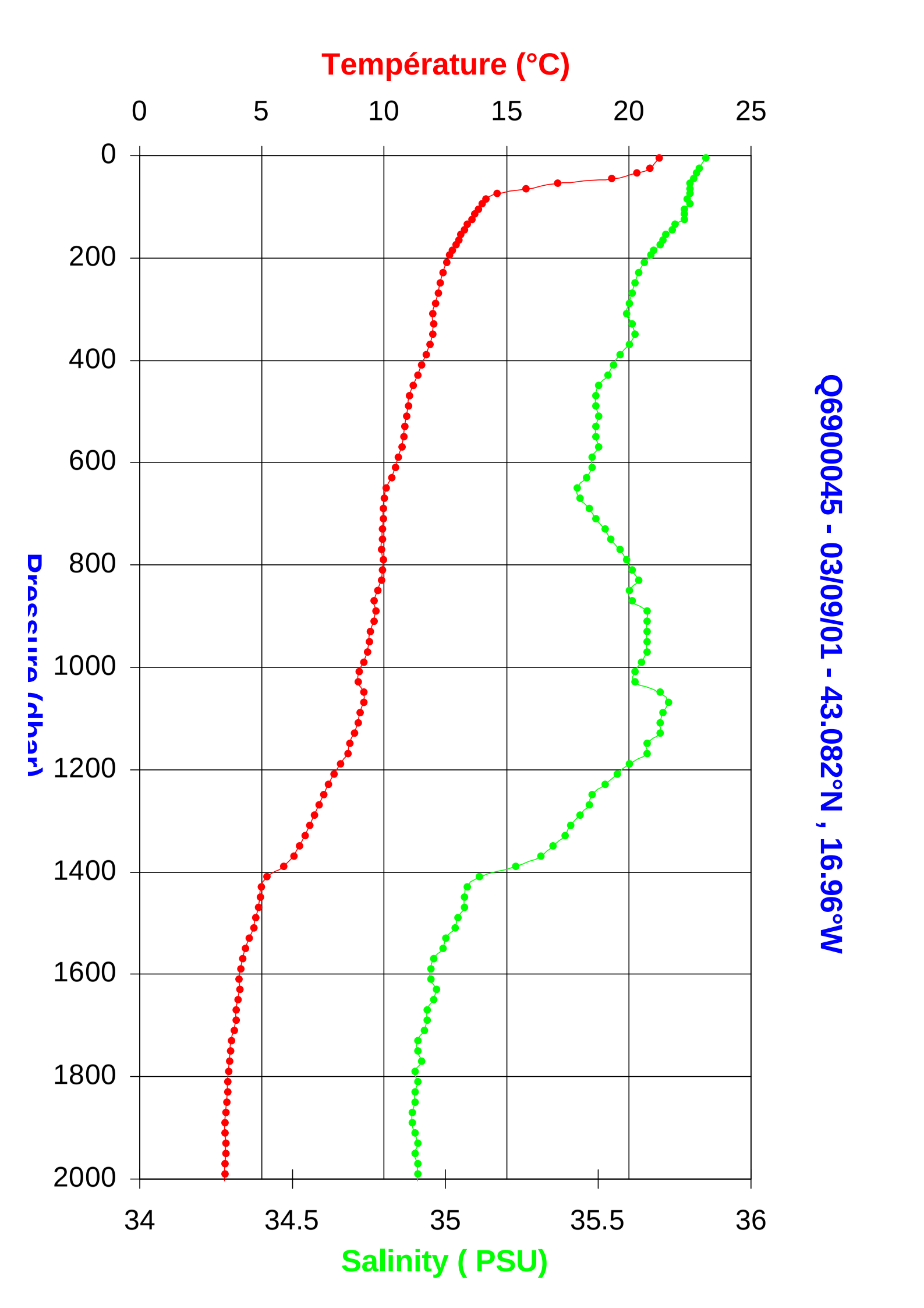
Maximum allowed error or number of points

More points when variations are higher

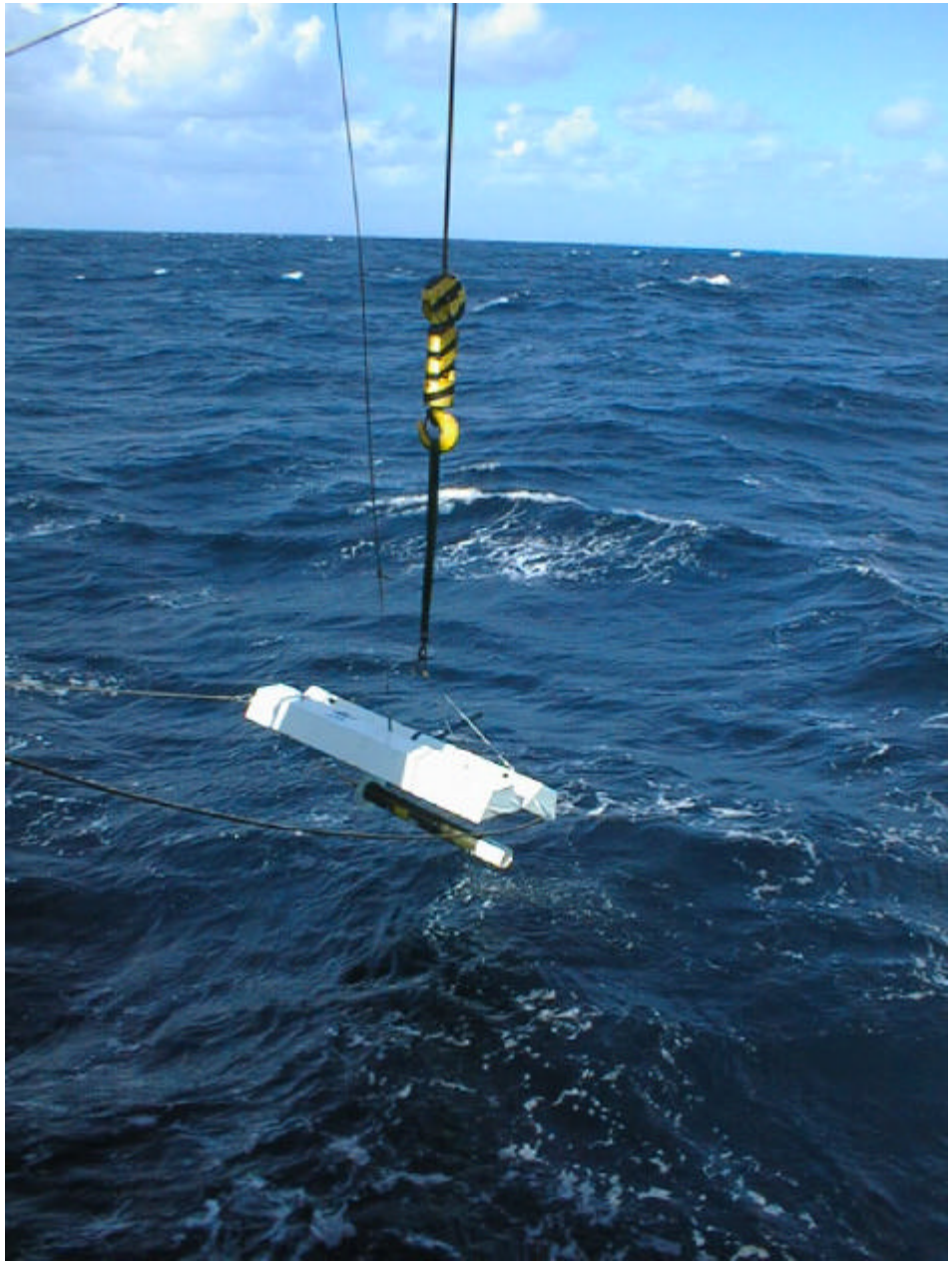


Data reduction : Slices Method

- 2 layers (upper and lower)
- each layer is divided into slices which height is user-defined
- all measurements which are gathered inside every slice are averaged (1 sample /meter)


Température (°C)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Pression (dbar)
34
34.5
35
35.5
36
Salinity (PSU)
Q6900045 - 03/09/01 - 43.082°N , 16.96°W

Provor deployments

- On going experiments (North Atlantic)
  - 1999-2000: technology experiment *Pommier* (5 profilers)
  - 2000-2001: scientific experiment *Pomme* (16 profilers including 6 Provor CT-F2)
- *GyroScope (funded by EU)* (40 profilers - PROVOR-CT-F2)
  - deployments in 2001 and 2002
- *Coriolis* (50 profilers - PROVOR-CT-F2)
  - deployments in 2002
- *Jamstec floats* : 70 METOCEAN PROVOR-CT-S
  - deployments in 2001 and 2002

Provor float deployment from a ship of the French Navy near the Azores